42588 - Data and data science

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Project 3: Data and Analysis

Purpose

The purpose of this project is to work with the analysis part of the PPDAC process, however, it is still relevant to consider how you communicate. The target audience consists of your fellow students who have followed the courses: introduction to statistics and introduction to machine learning.

Exercise 1 – Using the CAschools dataset

The overall question is how the average income in a district affects the average math score of students. To answer this you should use the data set "CAschools" without the observations for San Diego County and cover the tasks a)-e) below. For f)-g), you need to use the data for San Diego County as well.

a) Briefly **present** the question and data. **Calculate** relevant descriptive statistics for the main variables that you use in your analysis.
b) **Estimate** a linear regression model that captures the effect of average income on average math score. **Argue** why you include additional explanatory variables in your model besides average income. **Argue** why you prefer the model you see as your final model. **Interpret** the final regression model, i.e. comment on signs, significance and overall goodness-of-fit.
c) **Discuss** at least one interaction and one non-linear effect. It is OK if these are not included in the final linear regression model. However, in any case, you **need to argue** why they are included or not included.
d) **Make** at least one plot that you can use for model control. **Discuss** the validity of the model given your model control.
e) **Analyse** and **discuss** which of the explanatory variables that has the most influence on the dependent variable using elasticities or alternative measures.
f) **Analyse** and **discuss** how your results transfer to schools in San Diego County.
g) **Make** a conclusion as to how your m odel answers the overall question and discuss uncertainties in your answer.

See Appendix A for a description of the variables in the data.

Exercise 2 –Using your own data to answer a question

Here you need to formulate a question, where you can find data to answer the question. This exercise has the following guidelines

-The question has to be such that you need to apply a supervised method to answer the question. In addition, the question cannot be a pure prediction question. The answer to the question needs to depend on some interpretation of the supervised model that you apply. An example could be "How do car use and car age affect used car prices".

-The data have to consist of at least 250 observations. The data set has to include a dependent variable and at least one continuous variable and one discrete non-binary variable.

Given the question and data, your project needs to cover tasks a)-e) below

a) Briefly **present** the question and data. **Make** descriptive statistics for the main variables that you use in your analysis, i.e. describe the marginal distribution of each main variable and the association among these variables, e.g. using summary statistics and a correlation matrix.
b) **Estimate** a supervised model. **Test** if the continuous and discrete variables should enter linearly. **Test** at least one interaction effect.
c) **Argue** for a final model and i**nterpret** the final model, i.e. comment on signs, significance and overall goodness-of-fit. **Make** at least one plot that you can use for model control.
d) **Discuss** which of the explanatory variables that has the most influence on the dependent variable, and **calculate** the elasticity with respect to the continuous variables and some indicator for the effect of the discrete variables in the model.
e) **Make** a conclusion as to how your model answers the overall question and **discuss** uncertainties in your answer.

Exercises 3 – Car ownership

The overall question is how income affects car ownership. In this exercise, you should use the data "bilejerskab2025.xlsx", which is described below the questions.

a) **Find** the market shares for 0, 1 and 2 cars in the data. **Discuss** whether the market shares change between 2008 and 2009?
b) **Describe** how the number of cars relates to number of driver's licenses, income and number of kids, e.g. using cross tabulation, group averages or correlations. **Comment** on the results.
c) **Estimate** a logistic regression, where you model whether a family has a car or not. **Include** as many variables as possible to explain behaviour and consider non-linear effects. C**omment** on the estimated parameters and argue for your choice of final model.
d) **Estimate** a logistic regression modelling whether a car family has 1 or 2+ cars. C**omment** on the estimated parameters and argue why your choice of final model is the most suitable.
e) **Apply** the model(s) from c) and d) that you find necessary to assess how a 20 % increase in income affects the demand for cars, i.e. the probability of car or 2+ cars in one or both of you models. Alternatively, you are welcome to set up a logit model (or multinomial logistic regression) to answer this question instead. Apply your preferred approach to **predict** how many cars the sample is expected to have if each observation has a 20 % higher income. **Explain** your approach.
f) **Find** the elasticity of car ownership with respect to income according to your results in e)?
g) **Summarise** how income affects car ownership based on your results in a)-f).

In the data file, there are two sheets. The second sheet includes info on the various variables. The data include the variables:

Variable name	**Description**
ID	ID for each observation
HHLicences	No. driver's licenses
Gender	Gender
Children	No. children in household
Year	Year for interview
AgeGroup	Age group of respondent
LaborMarketAssociation	Employment info
Car	No. of cars
HHIncome	Household income
PerIncome	Person income

Exercises 4 – NOx pollution in Boston

The overall question here is how we can analyse whether an area in Boston has too high levels of NOx. We will use the data set Boston, which in R is included in the ISLR2 library. It will also be uploaded to DTU learn.

a) **Create** a dummy (*nox_dum*) for when the *nox* variable is above 0.60 parts per 10 million. **Calculate** descriptive statistics and **discuss** which variables are likely to explain *nox_dum* and whether you could get issues with multicollinearity.
b) **Estimate** a logistic regression to model *nox_dum* based on the other variables (use a significance level of 0.15 when deciding on variables). **Explain** how you do model selection and c**omment** on the estimation results.
c) **Estimate** an extended model where you **test** at least one non-linear effect and one interaction. **Decide** on a final model and **argue** why you prefer this as your final model.
d) **Apply** your model to produce two confusion matrices: one where you assign each observation to the most likely outcome with probability one; another where each observation is assigned to an outcome based on the model probability that this will be the outcome. **Discuss** the two matrices.
e) **Estimate** a linear regression to model *nox* based on the other relevant variables. **Compare** this model to your logistic regression considering that the goal is to find areas with NOx above the threshold.
f) Based on the above results and additional analyses that you find relevant, **discuss** which variables you find most important for the prediction of whether the NOx level is above the threshold 0.60 parts per 10 million.

See Appendix A for a description of the variables in the data.

Formalities

- The project should be created by the same groups as Projects 1 and 2 unless otherwise agreed with Stefan.
- The project report should be handed in the 8/5-2026 before 23.59 as pdf file on DTU Learn
- The final project report should be max. 20 pages (excl. appendices) and address the tasks mentioned below each of the exercises.
- The project report should include a section/table indicating the person(s) who has/have worked on the various parts of the report.
- The report should include a section indicating how you have used AI related to this project.
- Remember page numbers, table numbers, figure numbers, consistent references, etc.
- Remember that it is OK to move extra material to an appendix.
- The four exercises are weighted equally in the evaluation.
- The project is evaluated on a scale from 0 to 4 where 0 corresponds to not passed.
- Questions to all aspects of the exercise are welcome and will not affect the evaluation.

Appendix A

California Test Score Data

Description

The dataset contains data on test performance, school characteristics and student demographic backgrounds for school districts in California.

Usage

```
data("CASchools")
```

Format

A data frame containing 420 observations on 14 variables.

district

character. District code.

school

character. School name.

county

factor indicating county.

grades

factor indicating grade span of district.

students

Total enrollment.

teachers

Number of teachers.

calworks

Percent qualifying for CalWorks (income assistance).

lunch

Percent qualifying for reduced-price lunch.

computer

Number of computers.

expenditure

Expenditure per student.

income

District average income (in USD 1,000).

english

Percent of English learners.

read

Average reading score.

math

Average math score.

Details

The data used here are from all 420 K-6 and K-8 districts in California with data available for 1998 and 1999. Test scores are on the Stanford 9 standardized test administered to 5th grade students. School characteristics (averaged across the district) include enrollment, number of teachers (measured as "full-time equivalents", number of computers per classroom, and expenditures per student. Demographic variables for the students are averaged across the district. The demographic variables include the percentage of students in the public assistance program CalWorks (formerly AFDC), the percentage of students that qualify for a reduced price lunch, and the percentage of students that are English learners (that is, students for whom English is a second language).

Boston Data

Description

A data set containing housing values in 506 suburbs of Boston.

Usage

```
Boston
```

Format

A data frame with 506 rows and 13 variables.

`crim`

per capita crime rate by town.

`zn`

proportion of residential land zoned for lots over 25,000 sq.ft.

`indus`

proportion of non-retail business acres per town.

`chas`

Charles River dummy variable (= 1 if tract bounds river; 0 otherwise).

`nox`

nitrogen oxides concentration (parts per 10 million).

`rm`

average number of rooms per dwelling.

`age`

proportion of owner-occupied units built prior to 1940.

`dis`

weighted mean of distances to five Boston employment centres.

`rad`

index of accessibility to radial highways.

`tax`

full-value property-tax rate per $10,000.

`ptratio`

pupil-teacher ratio by town.

`lstat`

lower status of the population (percent).

`medv`

median value of owner-occupied homes in $1000s.